ASME IMECE® 2026
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RESEARCH PRESENTATION

Complex Spectral Binding Networks for High-Entropy Fluid Dynamics

Solving the Spectral Fidelity Paradox in Atmospheric Gravity Waves

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IMECE2026-192090

Gravity-wave reconstruction needs fine-scale structure

Physical role

Gravity waves transport momentum through the atmosphere.

Reconstruction challenge

Resolved momentum-flux fields contain irregular structure across multiple spatial scales.

Study focus

Compare how neural representations recover a masked region under the same model scale and training protocol.

pp. 1–2, Abstract and Section 1 (Introduction)

A controlled test of complex spectral binding

Which representation recovers the masked region at a matched model size?

01	**Method**	HONN with learned complex spectral weights
02	**Comparison**	Five architectures near 133K parameters; five seeds
03	**Evaluation**	Pointwise scores, power spectra, and field reconstructions

p. 2, Section 1 contribution list and Section 2.3.1; p. 3, Sections 2.4–2.5

Complex spectral binding inside HONN

Core idea: learned complex weights act on spectral features

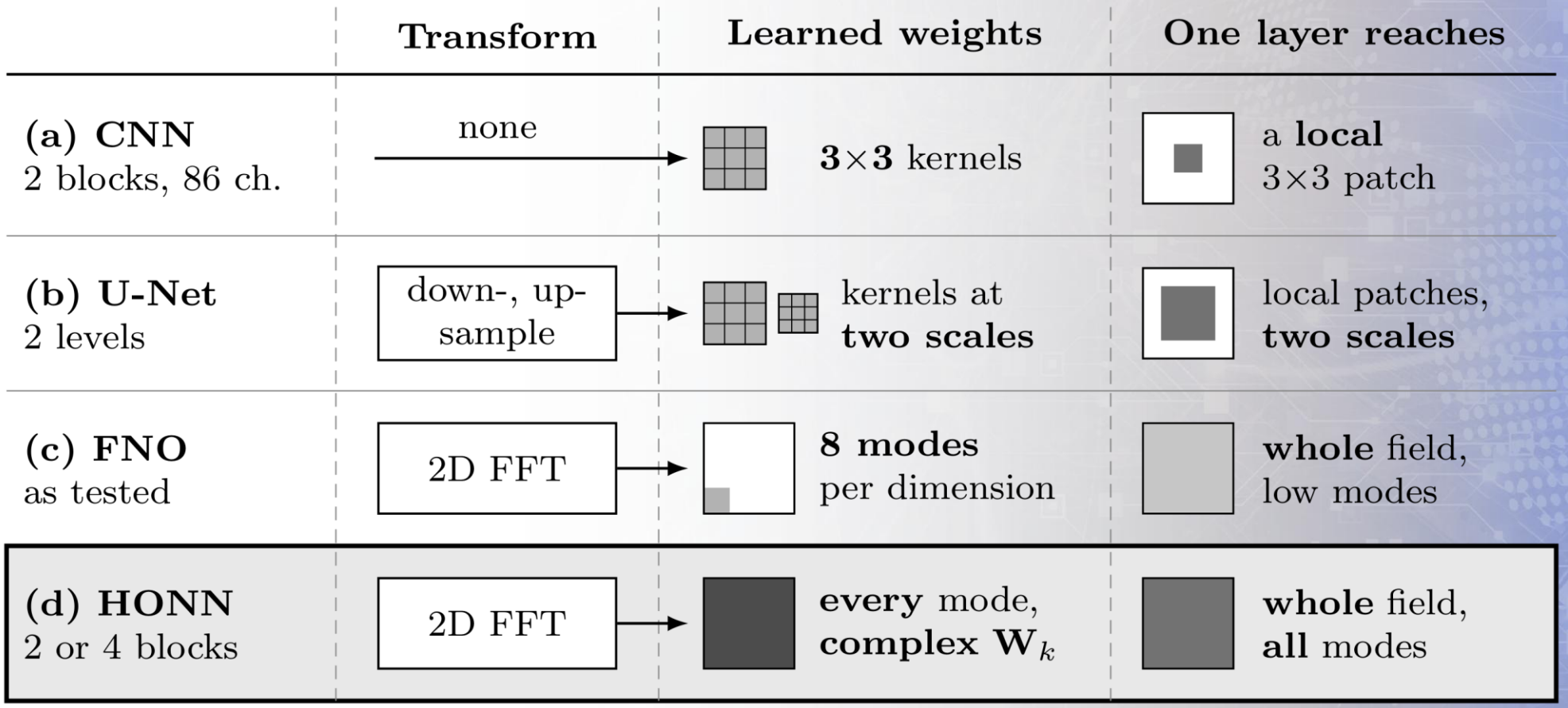


$$\mathbf{z}_{k+1} = \mathrm{ReLU}\left(\mathbf{z}_k + \mathcal{F}^{-1}[\,\mathbf{W}_k \odot \mathcal{F}(\mathbf{z}_k)\,]\right), \qquad \mathbf{W}_k \in \mathbb{C}^{C\times H\times W}$$

1 × 1 input lifting → repeated HoloBlocks → 1 × 1 output projection

p. 2, Section 2.3.1; p. 3, opening paragraph

HONN learns a complex weight for every frequency



pp. 2–3, Sections 2.3.1–2.3.4; p. 6, Section 4.2

Five models at a matched parameter budget

Model	Representation	Real parameters
HONN-2B	2 complex spectral blocks; 4 channels	133,253
HONN-4B	4 complex spectral blocks; 2 channels	133,199
CNN	Spatial convolution	135,279
U-Net	Multiscale encoder–decoder	131,977
FNO	Truncated spectral; 8 modes	131,665

Reported capacities are within ±2% of HONN-2B

pp. 2–3, Section 2.3 and Table 1

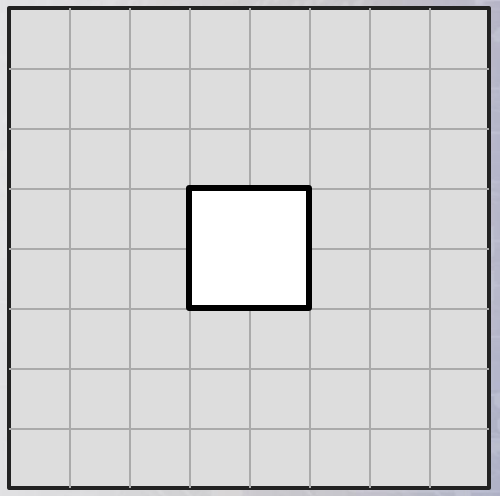
50,385 fields and one fixed center mask

50,385

momentum-flux fields

ECMWF IFS-1km

$$\rho_0 u' w' \quad (\mathrm{mPa})$$



128 × 128 field, central 32 × 32 masked

Training	Validation	Test
40,308	5,038	5,039

One fixed data split; five training seeds

p. 2, Sections 2.1–2.2; p. 3, Section 2.5

A shared training and evaluation protocol

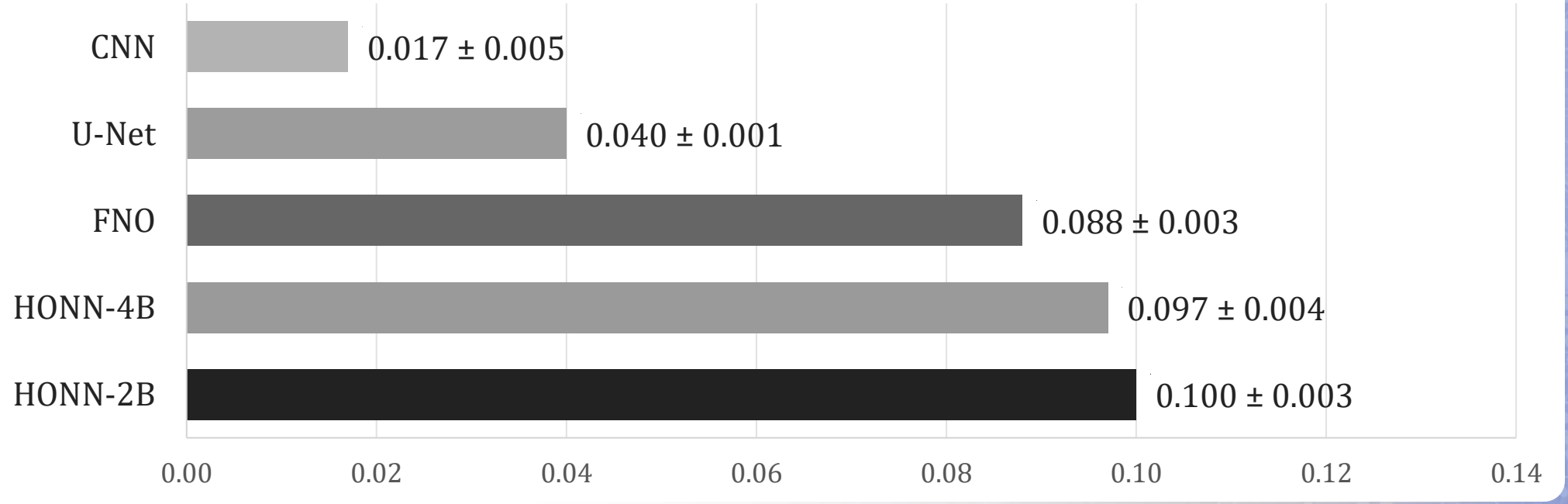
$$\mathcal{L} = \|\hat{\mathbf{y}} - \mathbf{y}\|_2^2 + 5\left\|(1 - \mathbf{M}) \odot (\hat{\mathbf{y}} - \mathbf{y})\right\|_2^2$$

Full-field error Extra masked-region penalty

Optimizer	Training budget	Evaluation
AdamW LR 10^{-3} Weight decay 10^{-4}	50 epochs Batch size 64 1,000 fields per epoch	Full validation split Full test split Five seeds

p. 2, Equation 1; p. 3, Section 2.4

HONN-2B leads the reported test R^2 comparison



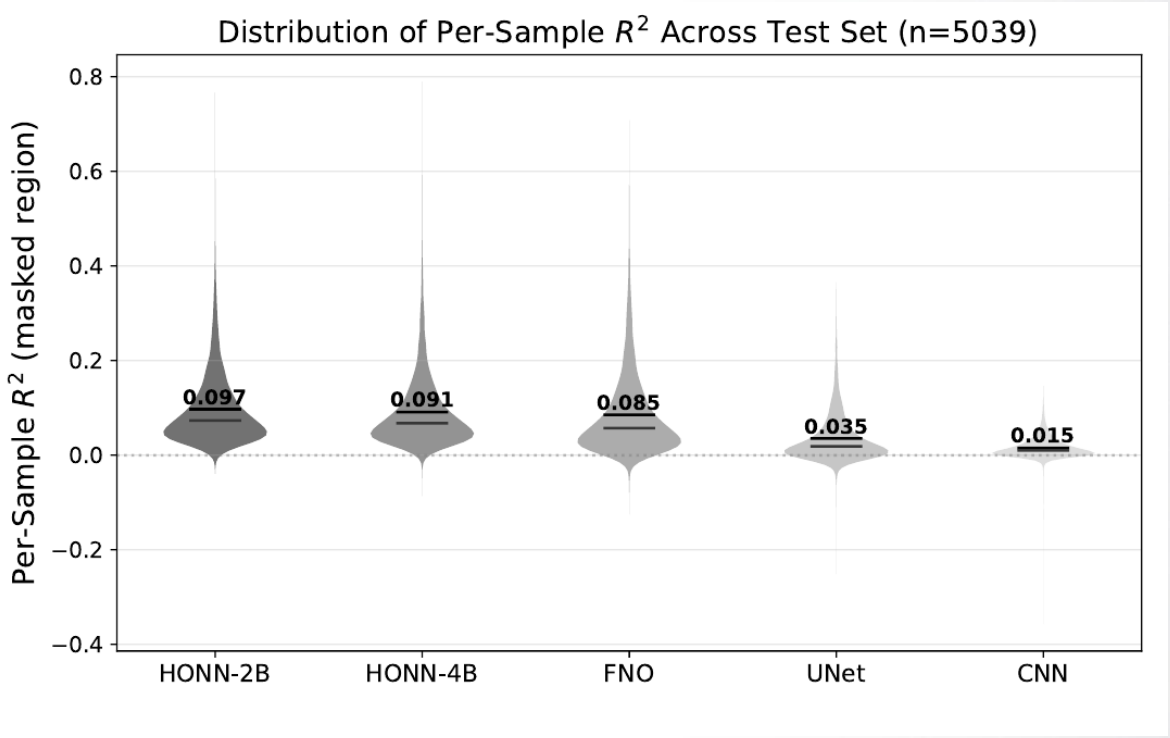
Mean ± SD, 5 seeds

vs FNO: p = 0.0004, Cohen's d = 4.13

vs HONN-4B: p = 0.189, not significant

p. 3, Section 3.1 and Table 2

HONN-2B has negative R^2 on only 3.2% of test fields



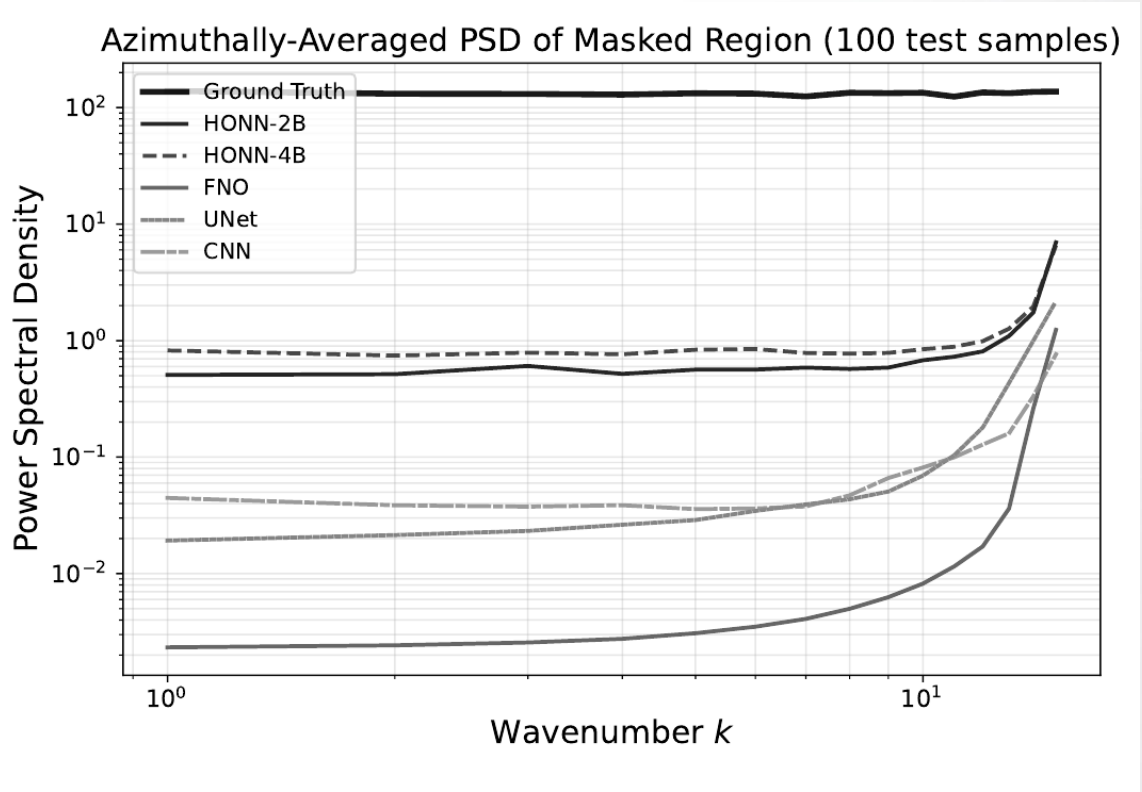
Per-field R^2 in the masked region, 5,039 test fields

Model	Mean R^2	Negative R^2
HONN-2B	**0.097**	**3.2%**
FNO	0.085	8.1%
U-Net	0.035	22%
CNN	0.015	18%

A separate statistic from the aggregate test R^2

p. 4, Section 3.4; p. 5, Figure 4

The metrics give different rankings



PSD: masked region, 100 test fields

Model	R^2 ↑	Spectral error ↓
HONN-2B	**0.100**	0.419
HONN-4B	0.097	0.477
FNO	0.088	0.344
U-Net	0.040	**0.305**
CNN	0.017	0.366

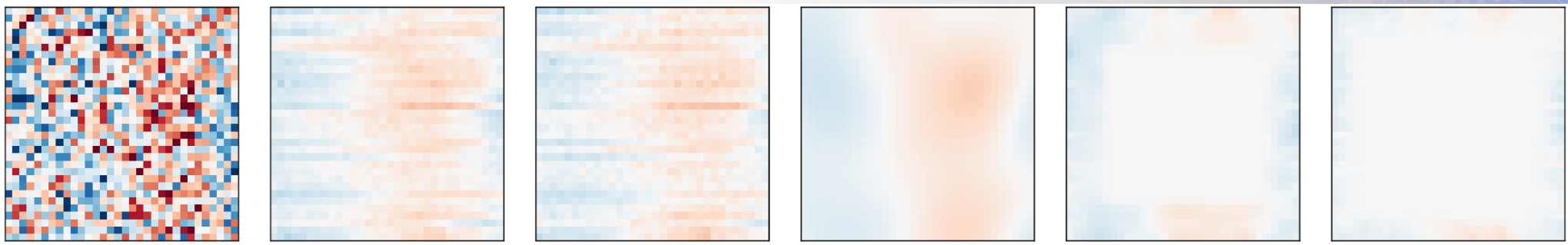
Different metrics select different models

p. 3, Table 2 and Section 2.5; p. 4, Figure 2 and Section 3.2

Sample 200: HONN fields retain visible texture

Of the two examples in the paper, the one with the higher HONN-2B R^2

Ground truth Target	**HONN-2B** **R^2 = 0.090**	HONN-4B R^2 = 0.082	FNO R^2 = 0.078	U-Net R^2 = 0.026	CNN R^2 = 0.008



HONN-2B has the highest R^2 among the five models on this sample.

Fine-scale target variation remains smoothed in all five reconstructions.

p. 5, Figure 3: Sample 200 (32 × 32 masked region), original paper colors

Conclusions

HONN-2B achieves the highest reported mean test R^2 in this benchmark.

Accuracy	HONN-2B: mean R^2 **0.100**, compared with 0.088 for FNO.
Architecture	HONN-2B vs HONN-4B: not significant ($p = 0.189$).
Field structure	Spectra and reconstructions show missing fine-scale detail.

p. 3, Table 2; pp. 4–6, results and discussion; p. 1, correspondence footnote

Prospects: isolate the mechanism and test broader use

Current comparison

Fixed 32 × 32 center mask

Approximately 133K parameters

Shared MSE training settings

Next experiments

Separate complex weights from spectral coverage

Scale model capacity

Test spectral losses and climate-model coupling

p. 6, Sections 4.2–4.3

Thank you

Questions & discussion

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